

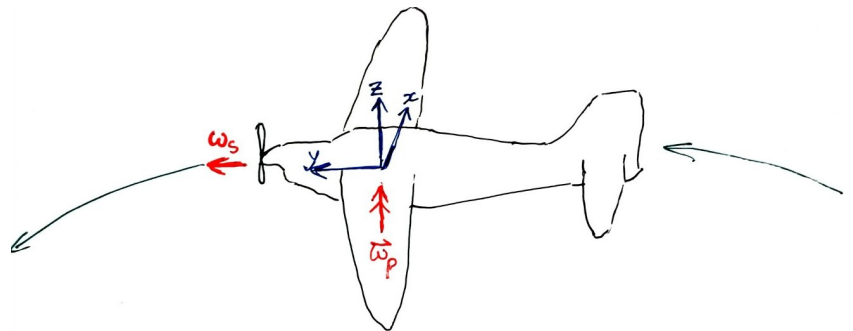
036026 Kinematics, Dynamics and Control of Robots

Homework 6 of 2026

This assignment is almost the same as that of the May 20, 2026 tutorial, with the only difference that numerical values are assigned for some parameters and variables, so that you can report numerical-valued vectors in the Moodle feedback tool. It is suggested that you first solve the problem symbolically (for instance, by using the MATLAB symbolics toolbox or Python's sympy module) and then, in the end, just substituting the given numerical values for the parameters and variables into your symbolic results and then calculating the required numerical-valued vectors.

An aircraft is flying in a horizontal circular arc, as depicted in the figure below. The figure shows the airframe axis frame, with the x axis pointing to the right of the aircraft, the y axis pointing forward and the z axis pointing upward. (When it is flying in an arc the airframe needs to be rolled to a small angle around the negative y airframe axis. For the purposes of this exercise you can ignore this small roll or bank angle.)

As a result of flying in this trajectory, the airframe experience a constant angular velocity $\vec{\omega}_p$ in the airframe z -axis direction. The propeller is driven by the engine at the constant speed ω_s about the centre line of the propeller shaft, which is perfectly aligned with the airframe y axis. In your analysis, in addition to the airframe frame, also use a body frame attached to the propeller such that the body frame y axis is parallel to the airframe y axis.



For the calculation of numerical-valued vectors, you should use the values $\omega_s = 420 \text{ rad/s}$ and $\omega_p = 0.1 \text{ rad/s}$.

You need to investigate the use of two different propellers:

- **Assignment part (a)** Consider a two-bladed propeller. You should assume that the body frame z axis is along one of the blades. With this and the y axis direction being known, the body frame x axis is defined. You should furthermore use the simplifying approximation that the combination of the two blades form a slender bar with a moment of inertia $I = 2.26 \text{ kgm}^2$ about both the body frame x and y axes, while the blade moment of inertia about the body z axis is zero. Of course the propeller body frame axis origin is located at the centre of mass of the bar.
- **Assignment part (b)** Consider an eight-bladed propeller. You should use the simplifying approximation that, for purposes of calculating the moment of inertia of the eight-bladed propeller, it can be approximated as a very thin disk, with the moment of inertia about a diametral axis of the disk of $I = 4.52 \text{ kgm}^2$ and about the axis normal to the disk through its centre of $2I$. With this assumption it is immaterial which orientation one uses for the x and

z axes of the body frame, as long as they are mutually orthogonal and as long as both lie in the plane of the disk. This means that the propeller body frame axis origin is located at the centre of mass of the disk.

For both parts (a) and (b) of the assignment you need to use the Newton-Euler equations as derived in MLS and determine the moment (torque) the propeller shaft is exerting on the propeller as a result of the described motion, at three different instances described below. In all cases ignore the torque of the engine and any aerodynamics. You need to calculate the torque as a vector, in each case reporting this vector expressed both in the propeller body frame and in the airframe frame (that is, you need to report on 12 vectors in total).

Consider an initial orientation of the propeller body frame as one where its three axes are in the same directions as the corresponding axes of the airframe frame. Let θ be the angle through which the body frame has rotated from the initial orientation, as the propeller spins (assuming that it was already spinning at the initial moment). You need to perform the analysis describe above for three cases:

- **Case 1** $\theta = 0$ radians.
- **Case 2** $\theta = \pi/4$ radians.
- **Case 3** $\theta = \pi/2$ radians.

Compare the torques of all three cases and in component forms in both frames and also of both propellers. Write a paragraph on your conclusions and/or observations. You need to report on Moodle on your findings.